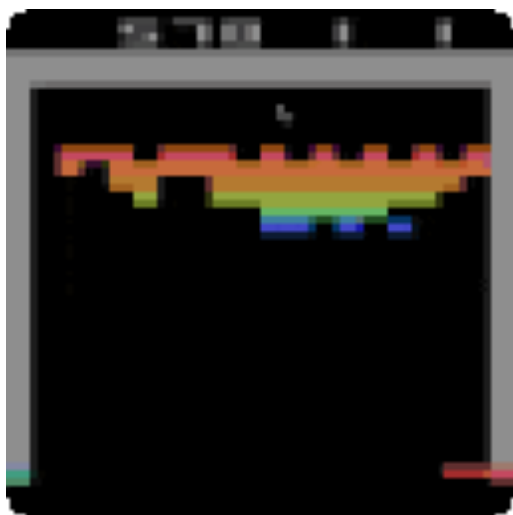




Sprint 3

03.06.26

Herausforderungen



- Kein informativer State
- Extrem viele zukünftige States
- Ein Bild enthält nicht alle notwendigen Infos
- Viele irrelevante Informationen
- Seltene Rewards



MuZero

Policy/Value Model

$$p^k, v^k = f_{\theta}(s^k)$$

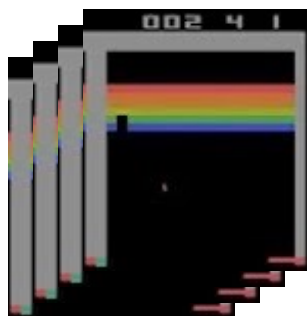
MCTS

Dynamics Model

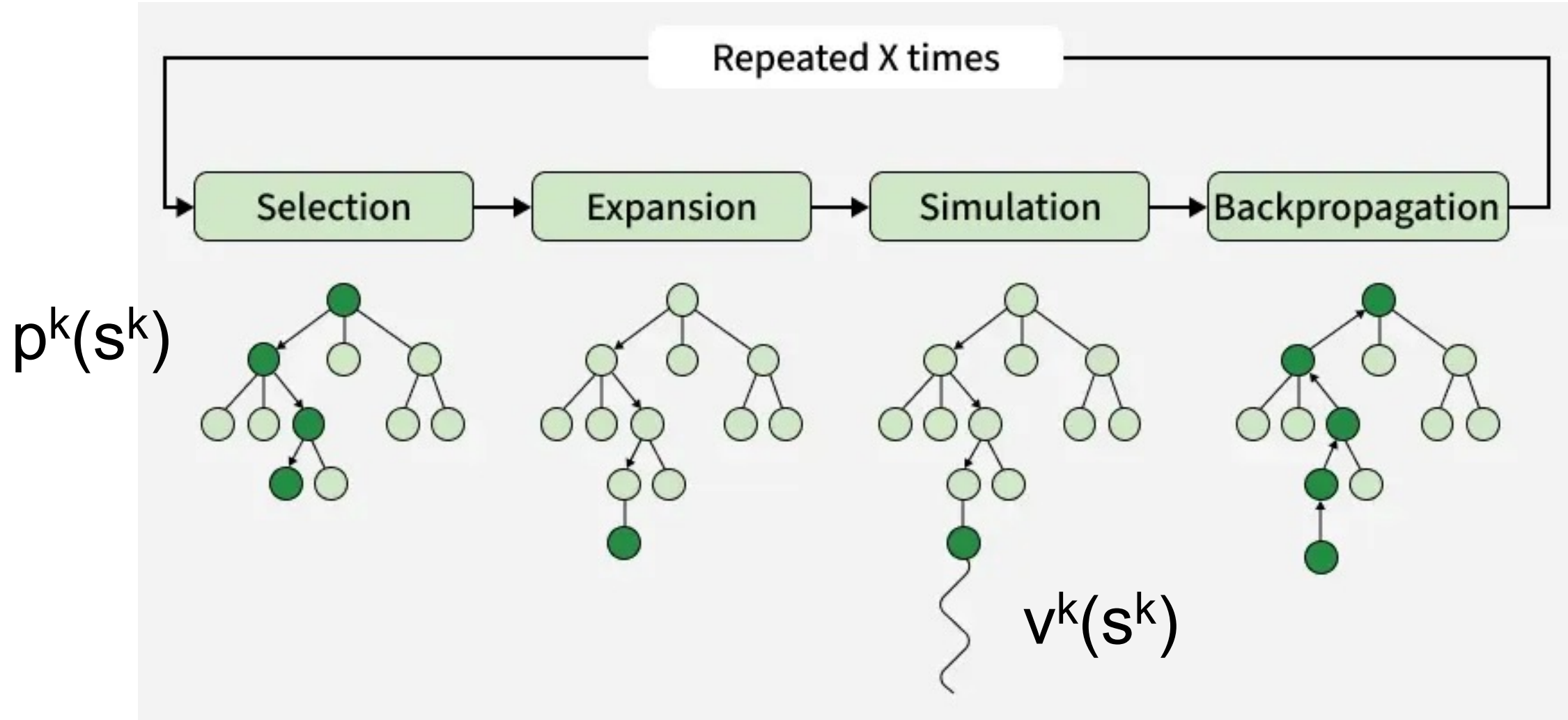
$$r^k, s^k = g_{\theta}(s^{k-1}, a^k)$$

Representation Model

$$s^0 = h_{\theta}(o_1, \dots, o_t)$$



t: step in environment
 o_t : observations
 k: step in MCTS
 s^0 : current root state
 a^k : action at step k
 s^k : state at step k
 r^k : reward at step k
 p^k : policy for MCTS
 v^k : value of state s^k





Ziele für Sprint 3

- Implementiert Policy/Value Model
- Folgt https://medium.com/@_michelangelo_/alphazero-for-dummies-5bcc713fc9c6
- CartPole und FrozenLake ohne (Leaf) Rollouts



Dreamer

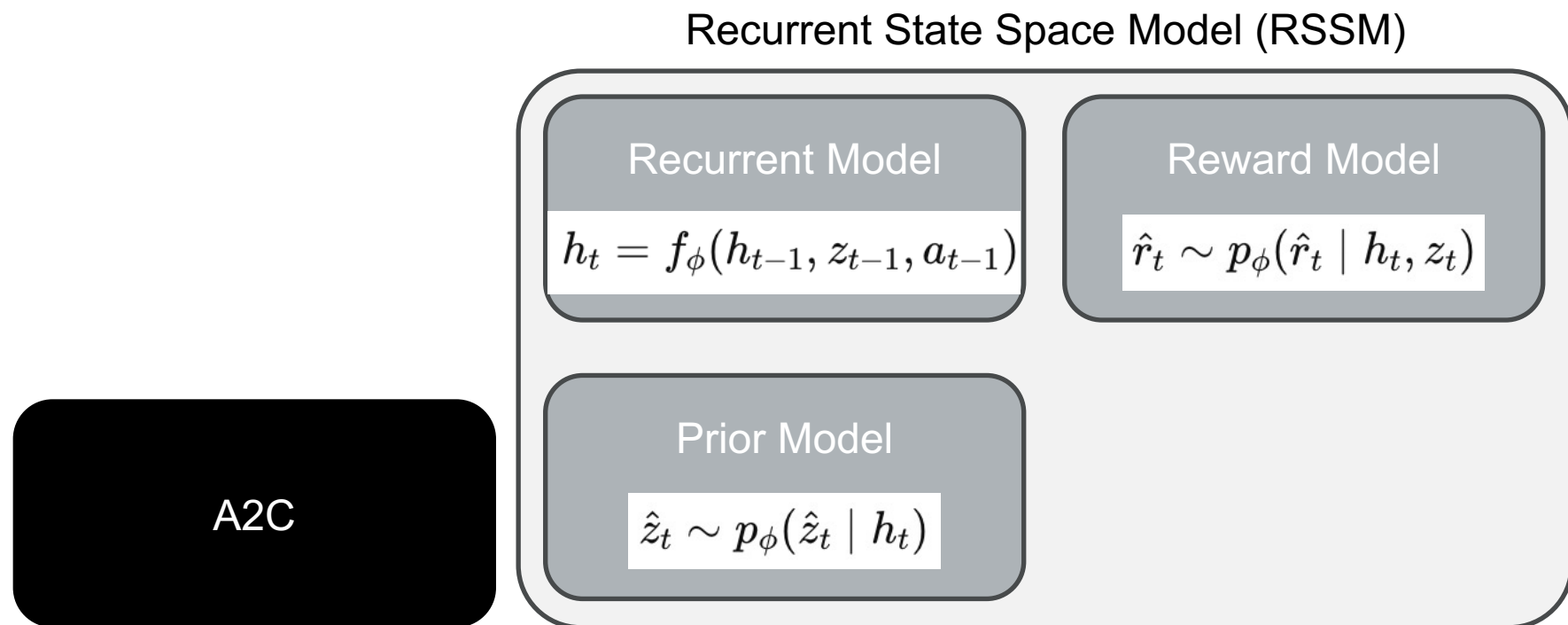
A2C

Dynamics Model

x_t : observation
 e_t : encoded observation
 h_t : recurrent state
 z_t : stochastic latent state
 a_t : action
 r_t : reward



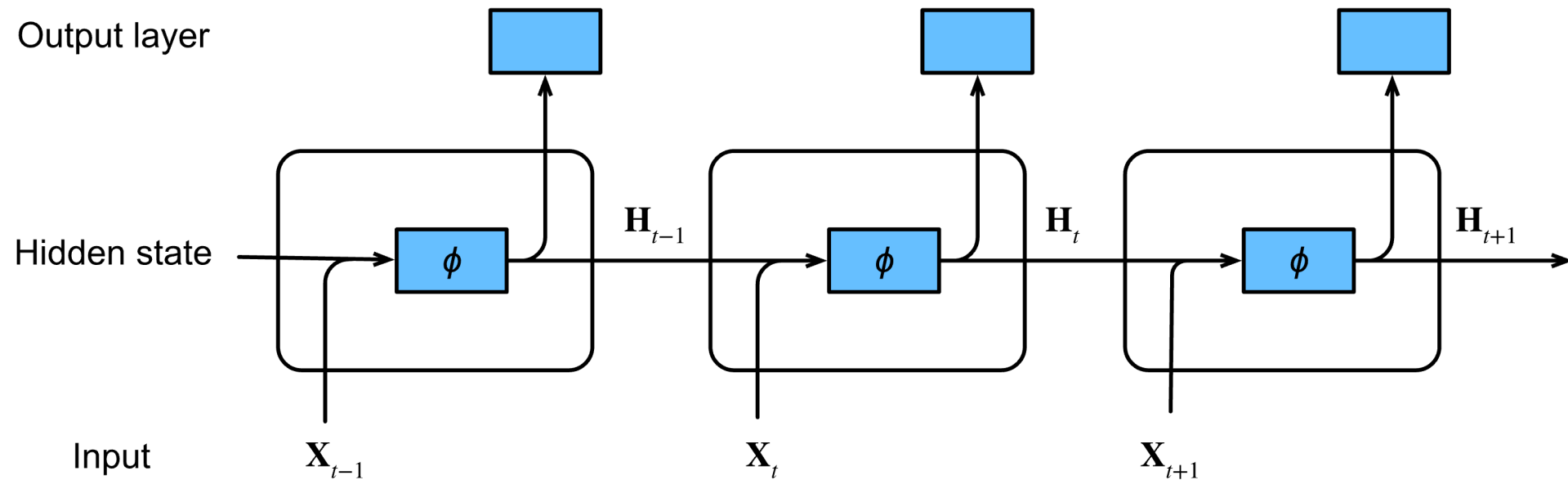
Dreamer



x_t : observation
 e_t : encoded observation
 h_t : recurrent state
 z_t : stochastic latent state
 a_t : action
 r_t : reward



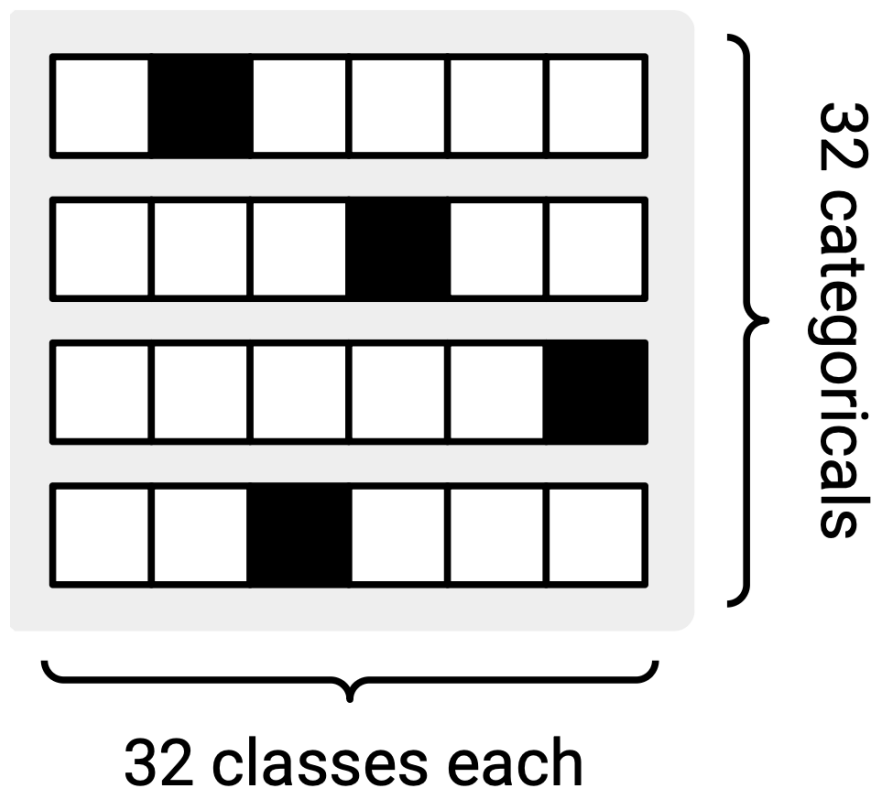
Recurrent State h_t



Gated recurrent unit (GRU)



Stochastic latent state z_t



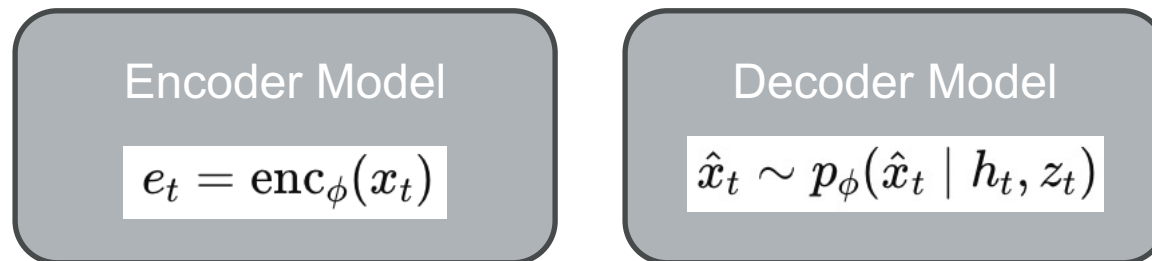
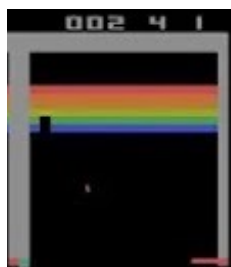
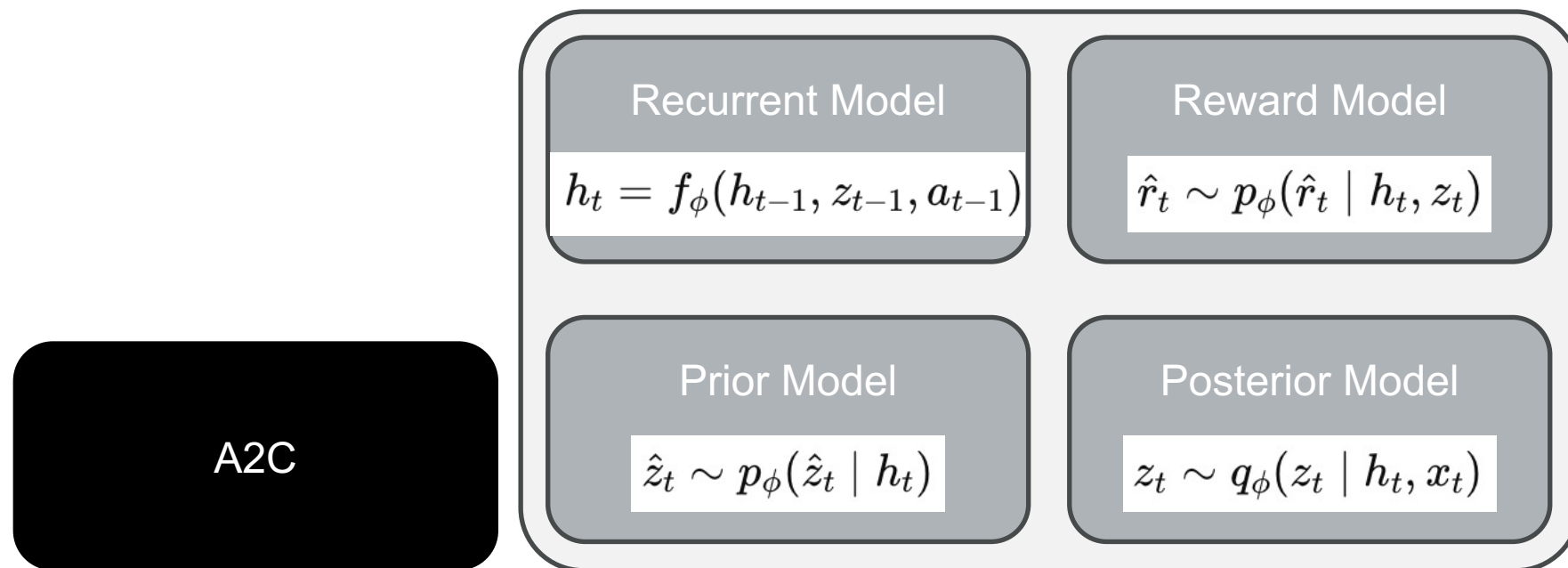
- Prior Model output: 32 x 32 Wahrscheinlichkeitswerte
- z_t ist ein Sample





Dreamer

Recurrent State Space Model (RSSM)



x_t : observation
 e_t : encoded observation
 h_t : recurrent state
 z_t : stochastic latent state
 a_t : action
 r_t : reward



Ziele für Sprint 3

- Implementiert RSSM
- Folgt Anweisungen im PDF
- Halbwegs didaktisches Tutorial:
<https://medium.com/%40lukasbierling/recurrent-state-space-models-pytorch-implementation-ba5d7e063d11>
- CartPole ohne Geschwindigkeitsinformationen